

EE 101: Basic Electronics

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Lecture Plan

- ① Polyphase circuits
 - Balanced three phase system
 - $Y - \Delta$ connection
 - Three phase power measurement
- ② Magnetic circuits
- ③ Filter circuits and resonance

Polyphase circuits

Polyphase Circuits

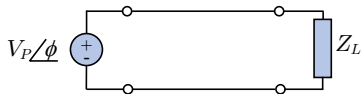


Figure: Single-phase system

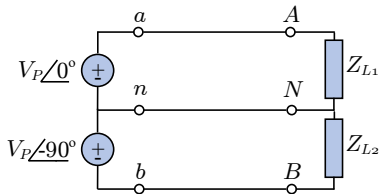


Figure: Two-phase three-wire system

Circuits or systems in which the ac sources operate at the same frequency but different phases are known as polyphase.

Three-phase Transmission Lines over IIT Guwahati Campus



Three-phase Lines

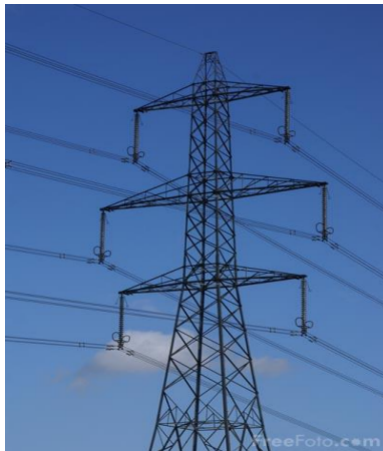


Figure: 3-phase Transmission line

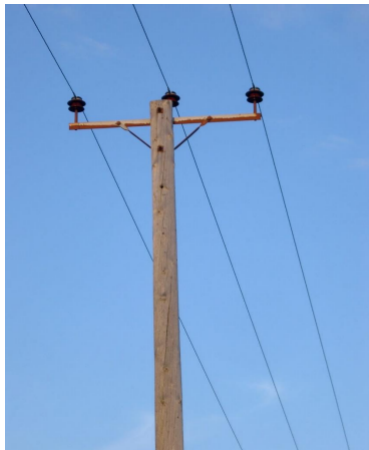


Figure: 3-phase distribution line

Three Phase System

- AC systems came into existence in the late 1880s because of the generation, transmission and consumption limitation of DC power .
- During the first use of AC, one basic problem remained: single phase motors are not able to start by themselves. This led to the invention of poly phase system in the early 1890s.



Figure: Motor

Three Phase System (cont..)

Consequently some advantages of 3-phase system over 1-phase system came out to be:

- 3-phase generators, motors and transformers are simpler, cheaper and more efficient.
- The instantaneous power in a three-phase system can be constant (not pulsating). This results in uniform power transmission and less vibration of three-phase machines.
- 3-phase transmission lines deliver more power for a given cost or for a given weight of conductor.
- Voltage regulation of a 3-phase system is inherently better.

Three Phase System (cont..)

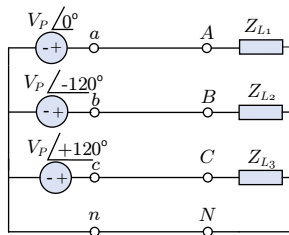
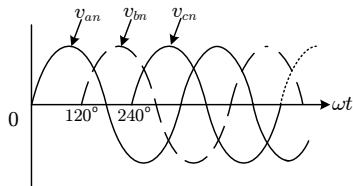
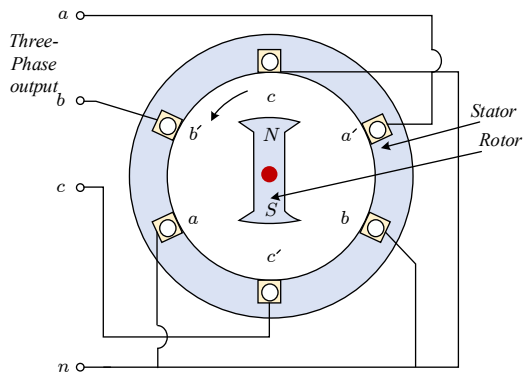


Figure: Three-phase four-wire system

Three Phase System (cont..)

$$v_{an} = \sqrt{2}V_p \cos \omega t,$$

$$v_{bn} = \sqrt{2}V_p \cos(\omega t - 120^\circ)$$

$$v_{cn} = \sqrt{2}V_p \cos(\omega t + 120^\circ)$$

Note:

$$v_{an} + v_{bn} + v_{cn} = 0$$

V_p is rms value of the phase voltages

Phasor representation of the equations are

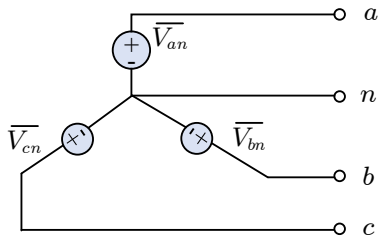
$$\overline{V_{an}} = V_p \underline{\angle 0^\circ} = |\overline{V_{an}}| \underline{\angle 0^\circ}$$

$$\overline{V_{bn}} = V_p \underline{\angle -120^\circ} = |\overline{V_{bn}}| \underline{\angle -120^\circ}$$

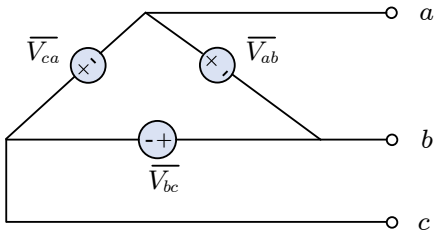
$$\overline{V_{cn}} = V_p \underline{\angle +120^\circ} = |\overline{V_{cn}}| \underline{\angle +120^\circ}$$

$$= V_p \underline{\angle -240^\circ} = |\overline{V_{cn}}| \underline{\angle -240^\circ}$$

Y and Δ Connections



(a)



(b)

Figure: Three-phase voltage sources: (a) Y-connected source, (b) Δ connected source

Balanced Three Phase Voltages

$$\overline{V_{an}} + \overline{V_{bn}} + \overline{V_{cn}} = 0$$

$$|\overline{V_{an}}| = |\overline{V_{bn}}| = |\overline{V_{cn}}|$$

$$\begin{aligned}\overline{V_{an}} + \overline{V_{bn}} + \overline{V_{cn}} &= V_p \angle 0^\circ + V_p \angle -120^\circ + V_p \angle +120^\circ \\ &= V_p (1 - 0.5 - j0.866 - 0.5 + j0.866) \\ &= 0\end{aligned}$$

Balanced phase voltages are equal in magnitude and are shifted from each other by 120° .

Phase Sequence

Positive phase sequence (Anti-clockwise rotation of rotor)

$$\overline{V_{an}} = V_p \angle 0^\circ$$

$$\overline{V_{bn}} = V_p \angle -120^\circ$$

$$\overline{V_{cn}} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

Negative phase sequence (Clockwise rotation of rotor)

$$\overline{V_{an}} = V_p \angle 0^\circ$$

$$\overline{V_{cn}} = V_p \angle -120^\circ$$

$$\overline{V_{bn}} = V_p \angle -240^\circ = V_p \angle +120^\circ$$

The phase sequence is the time order in which the voltage pass through their respective maximum values.

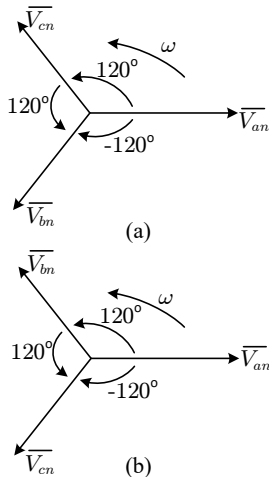


Figure: Phase sequences: (a) abc or positive sequence, (b) acb or negative sequence

Balanced Three Phase Loads

A balanced load is one in which the phase impedances are equal in magnitude and in phase.

$$\bar{Z}_1 = \bar{Z}_2 = \bar{Z}_3 = \bar{Z}_Y$$

$$\bar{Z}_a = \bar{Z}_b = \bar{Z}_c = \bar{Z}_\Delta$$

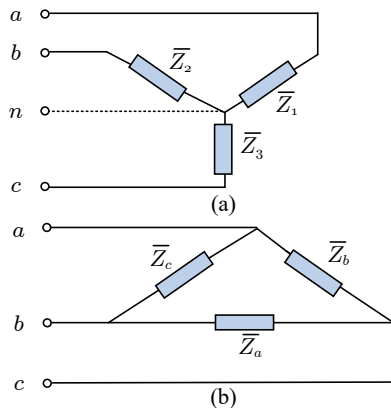


Figure: Two possible three-phase load configurations: (a) Y-connected
(b) Δ -connected

Δ - Y conversion

For circuit analysis, the star connected loads can be represented using the delta connected loads and vice-versa.

The impedance seen between the lines a and b in the star connected load is $\bar{Z}_1 + \bar{Z}_2$ (series combination of $\bar{Z}_1$ and $\bar{Z}_2$). In the delta connected load, the impedance seen between a and b is $\bar{Z}_c$ in parallel with the series combination of $\bar{Z}_b$ and $\bar{Z}_a$

$$\begin{aligned}\bar{Z}_1 + \bar{Z}_2 &= \bar{Z}_c \parallel (\bar{Z}_a + \bar{Z}_b) \\ &= \frac{\bar{Z}_c(\bar{Z}_a + \bar{Z}_b)}{\bar{Z}_a + \bar{Z}_b + \bar{Z}_c}\end{aligned}$$

Similarly,

$$\bar{Z}_2 + \bar{Z}_3 = \frac{\bar{Z}_a(\bar{Z}_c + \bar{Z}_b)}{\bar{Z}_a + \bar{Z}_b + \bar{Z}_c}$$

$$\bar{Z}_3 + \bar{Z}_1 = \frac{\bar{Z}_b(\bar{Z}_c + \bar{Z}_a)}{\bar{Z}_a + \bar{Z}_b + \bar{Z}_c}$$

Δ - Y conversion (cont..)

Solving these three equations, the star connected impedances can be represented with equivalent delta connected impedances as

$$\overline{Z}_1 = \frac{\overline{Z}_c \overline{Z}_b}{\overline{Z}_a + \overline{Z}_b + \overline{Z}_c}$$

$$\overline{Z}_2 = \frac{\overline{Z}_c \overline{Z}_a}{\overline{Z}_a + \overline{Z}_b + \overline{Z}_c}$$

$$\overline{Z}_3 = \frac{\overline{Z}_a \overline{Z}_b}{\overline{Z}_a + \overline{Z}_b + \overline{Z}_c}$$

Y - Δ conversion (cont..)

Similarly, delta connected loads ($\overline{Z}_a$, $\overline{Z}_b$ and $\overline{Z}_c$) can be derived in terms of star connected loads ($\overline{Z}_1$, $\overline{Z}_2$ and $\overline{Z}_3$).

$$\overline{Z}_a = \frac{\overline{Z}_1 \overline{Z}_2 + \overline{Z}_2 \overline{Z}_3 + \overline{Z}_3 \overline{Z}_1}{\overline{Z}_1}$$

$$\overline{Z}_b = \frac{\overline{Z}_1 \overline{Z}_2 + \overline{Z}_2 \overline{Z}_3 + \overline{Z}_3 \overline{Z}_1}{\overline{Z}_2}$$

$$\overline{Z}_c = \frac{\overline{Z}_1 \overline{Z}_2 + \overline{Z}_2 \overline{Z}_3 + \overline{Z}_3 \overline{Z}_1}{\overline{Z}_3}$$

Y - Δ conversion for Balanced Loads

If loads are balanced then

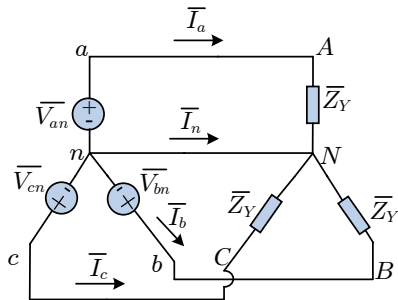
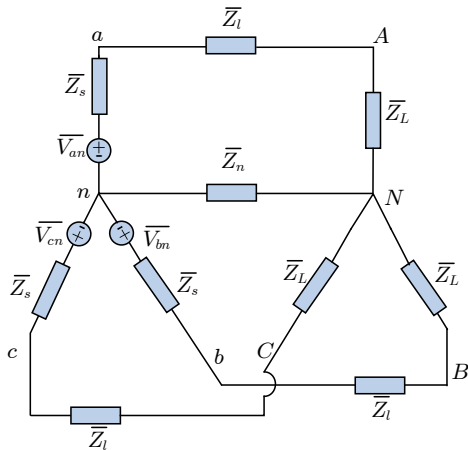
$$\overline{Z}_{\Delta} = 3\overline{Z}_Y \text{ or } \overline{Z}_Y = \frac{1}{3}\overline{Z}_{\Delta}$$

Y- Δ connections

Since both the three-phase source and the three-phase load can be either wye- or delta-connected, we have four possible connections:

- Y-Y connection (i.e., Y-connected source with a Y-connected load).
- Y- Δ connection.
- Δ - Δ connection.
- Δ -Y connection.

Y-Y system



A balanced Y-Y system, showing the source, line, and load impedances

$$\bar{Z}_Y = \bar{Z}_s + \bar{Z}_l + \bar{Z}_L$$

Y-Y system (cont..)

$$\overline{V_{an}} = V_p \angle 0^\circ$$

$$\overline{V_{bn}} = V_p \angle -120^\circ, \quad \overline{V_{cn}} = V_p \angle +120^\circ$$

$$\begin{aligned}\overline{V_{ab}} &= \overline{V_{an}} + \overline{V_{nb}} = \overline{V_{an}} - \overline{V_{bn}} \\ &= V_p \angle 0^\circ - V_p \angle -120^\circ \\ &= V_p \left(1 + \frac{1}{2} + j \frac{\sqrt{3}}{2} \right) = \sqrt{3} V_p \angle 30^\circ\end{aligned}$$

$$\overline{V_{bc}} = \overline{V_{bn}} - \overline{V_{cn}} = \sqrt{3} V_p \angle -90^\circ$$

$$\overline{V_{ca}} = \overline{V_{cn}} - \overline{V_{an}} = \sqrt{3} V_p \angle -210^\circ$$

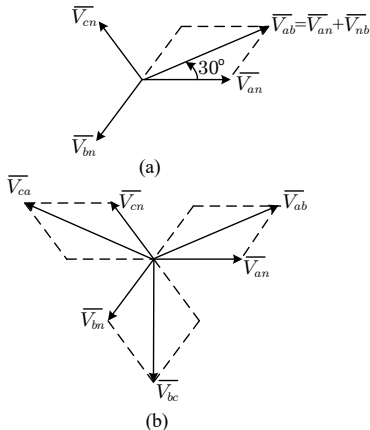


Figure: Phasor diagrams illustrating the relationship between line voltages and phase voltages

Y-Y system (cont..)

$$V_L = \sqrt{3}V_p$$

$$V_p = |\overline{V_{an}}| = |\overline{V_{bn}}| = |\overline{V_{cn}}|$$

$$V_L = |\overline{V_{ab}}| = |\overline{V_{bc}}| = |\overline{V_{ca}}|$$

Y-Y system (cont..)

$$\begin{aligned}\bar{I}_a &= \frac{\bar{V}_{an}}{\bar{Z}_Y}, \bar{I}_b = \frac{\bar{V}_{bn}}{\bar{Z}_Y} = \frac{\bar{V}_{an}/-120^\circ}{\bar{Z}_Y} \\ &= \bar{I}_a/-120^\circ \\ \bar{I}_c &= \frac{\bar{V}_{cn}}{\bar{Z}_Y} = \frac{\bar{V}_{an}/-240^\circ}{\bar{Z}_Y} \\ &= \bar{I}_a/-240^\circ\end{aligned}$$

$$\bar{I}_a + \bar{I}_b + \bar{I}_c = 0$$

$$\bar{I}_n = -(\bar{I}_a + \bar{I}_b + \bar{I}_c) = 0$$

$$\bar{V}_{nN} = \bar{Z}_n \bar{I}_n = 0$$

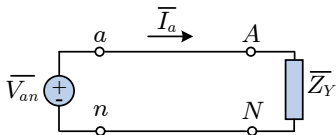


Figure: A single-phase equivalent circuit

$$\bar{I}_a = \frac{\bar{V}_{an}}{\bar{Z}_Y}$$

Y-Δ system

$$\overline{V_{an}} = V_p \angle 0^\circ$$

$$\overline{V_{bn}} = V_p \angle -120^\circ, \quad \overline{V_{cn}} = V_p \angle +120^\circ$$

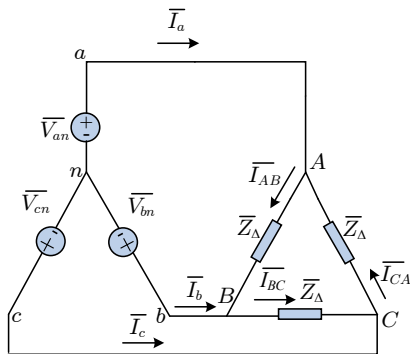
$$\overline{V_{ab}} = \sqrt{3}V_p \angle 30^\circ = \overline{V_{AB}},$$

$$\overline{V_{bc}} = \sqrt{3}V_p \angle -90^\circ = \overline{V_{BC}}$$

$$\overline{V_{ca}} = \sqrt{3}V_p \angle -150^\circ = \overline{V_{CA}}$$

$$\overline{I_{AB}} = \frac{\overline{V_{AB}}}{\overline{Z_{\Delta}}}, \quad \overline{I_{BC}} = \frac{\overline{V_{BC}}}{\overline{Z_{\Delta}}},$$

$$\overline{I_{CA}} = \frac{\overline{V_{CA}}}{\overline{Z_{\Delta}}}$$



$$-\overline{V_{an}} + \overline{Z_{\Delta}} \overline{I_{AB}} + \overline{V_{bn}} = 0$$

$$\overline{I_{AB}} = \frac{\overline{V_{an}} - \overline{V_{bn}}}{\overline{Z_{\Delta}}} = \frac{\overline{V_{ab}}}{\overline{Z_{\Delta}}} = \frac{\overline{V_{AB}}}{\overline{Z_{\Delta}}}$$

Y-Δ system (cont..)

$$\bar{I}_a = \bar{I}_{AB} - \bar{I}_{CA}, \quad \bar{I}_b = \bar{I}_{BC} - \bar{I}_{AB},$$

$$\bar{I}_c = \bar{I}_{CA} - \bar{I}_{BC}$$

$$\text{Since } \bar{I}_{CA} = \bar{I}_{AB} \angle -240^\circ,$$

$$\begin{aligned}\bar{I}_a &= \bar{I}_{AB} - \bar{I}_{CA} = \bar{I}_{AB}(1 - 1 \angle -240^\circ) \\ &= \bar{I}_{AB}(1 + 0.5 - j0.866) \\ &= \bar{I}_{AB}\sqrt{3} \angle -30^\circ\end{aligned}$$

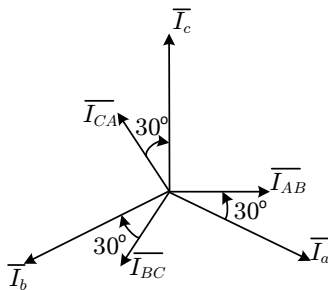


Figure: Phasor diagram illustrating the relationship between phase and line currents

$$I_L = \sqrt{3}I_P$$

Y- Δ system (cont..)

$$|\bar{I}_L| = |\bar{I}_a| = |\bar{I}_b| = |\bar{I}_c|$$

$$|\bar{I}_P| = |\bar{I}_{AB}| = |\bar{I}_{BC}| = |\bar{I}_{CA}|$$

$$\bar{Z}_Y = \frac{\bar{Z}_\Delta}{3}$$

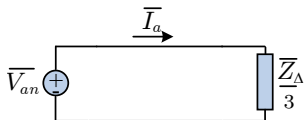


Figure: A single-phase equivalent circuit of a balanced circuit

Δ - Δ system

$$\overline{V}_{ab} = V_p \angle 0^\circ$$

$$\overline{V}_{bc} = V_p \angle -120^\circ, \quad \overline{V}_{ca} = V_p \angle +120^\circ$$

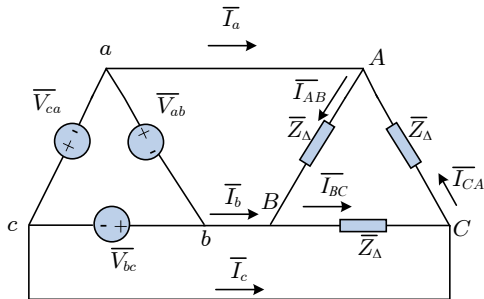
$$\overline{V}_{ab} = \overline{V}_{AB}, \quad \overline{V}_{bc} = \overline{V}_{BC},$$

$$\overline{V}_{ca} = \overline{V}_{CA}$$

$$\overline{I}_{AB} = \frac{\overline{V}_{AB}}{\overline{Z}_{\Delta}} = \frac{\overline{V}_{ab}}{\overline{Z}_{\Delta}},$$

$$\overline{I}_{BC} = \frac{\overline{V}_{BC}}{\overline{Z}_{\Delta}} = \frac{\overline{V}_{bc}}{\overline{Z}_{\Delta}},$$

$$\overline{I}_{CA} = \frac{\overline{V}_{CA}}{\overline{Z}_{\Delta}} = \frac{\overline{V}_{ca}}{\overline{Z}_{\Delta}}$$



$$\overline{I}_a = \overline{I}_{AB} - \overline{I}_{CA}, \quad \overline{I}_b = \overline{I}_{BC} - \overline{I}_{AB},$$

$$\overline{I}_c = \overline{I}_{CA} - \overline{I}_{BC}$$

$$I_L = \sqrt{3} I_P$$

Δ -Y system

$$\bar{V}_{ab} = V_p \underline{0^\circ}, \quad \bar{V}_{bc} = V_p \underline{-120^\circ}$$

$$\bar{V}_{ca} = V_p \underline{+120^\circ}$$

$$-\bar{V}_{ab} + \bar{Z}_Y \bar{I}_a - \bar{Z}_Y \bar{I}_b = 0$$

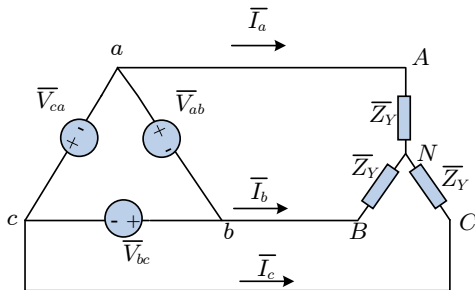
$$\bar{Z}_Y (\bar{I}_a - \bar{I}_b) = \bar{V}_{ab} = V_p \underline{0^\circ}$$

$$\bar{I}_a - \bar{I}_b = \frac{V_p \underline{0^\circ}}{\bar{Z}_Y}$$

$$\bar{I}_a - \bar{I}_b = \bar{I}_a (1 - 1 \underline{-120^\circ})$$

$$= \bar{I}_a \left(1 + \frac{1}{2} + j \frac{\sqrt{3}}{2}\right) = \bar{I}_a \sqrt{3} \underline{30^\circ}$$

$$\bar{I}_a = \frac{V_p \underline{-30^\circ}}{\bar{Z}_Y}$$



Δ -Y Transformation of Source

$$\overline{V}_{an} = \frac{V_p}{\sqrt{3}} \angle -30^\circ,$$

$$\overline{V}_{bn} = \frac{V_p}{\sqrt{3}} \angle -150^\circ, \quad \overline{V}_{cn} = \frac{V_p}{\sqrt{3}} \angle +90^\circ$$

$$\overline{I}_a = \frac{V_p \angle -30^\circ}{\sqrt{3} \overline{Z}_Y}$$

$$\overline{V}_{AN} = \overline{I}_a \overline{Z}_Y = \frac{V_p}{\sqrt{3}} \angle -30^\circ$$

$$\overline{V}_{BN} = \overline{V}_{AN} \angle -120^\circ,$$

$$\overline{V}_{CN} = \overline{V}_{AN} \angle +120^\circ$$

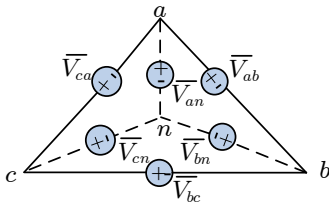


Figure: Transforming a Δ -connected source to an equivalent Y-connected source

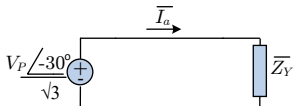


Figure: Transforming a Δ -connected source to an equivalent Y-connected source

Unbalanced Three Phase Systems

An unbalanced system is due to unbalanced voltage sources or an unbalanced load.

$$\bar{I}_a = \frac{\bar{V}_{AN}}{\bar{Z}_A}, \bar{I}_b = \frac{\bar{V}_{BN}}{\bar{Z}_B}, \bar{I}_c = \frac{\bar{V}_{CN}}{\bar{Z}_C}$$

$$\bar{I}_n = -(\bar{I}_a + \bar{I}_b + \bar{I}_c)$$

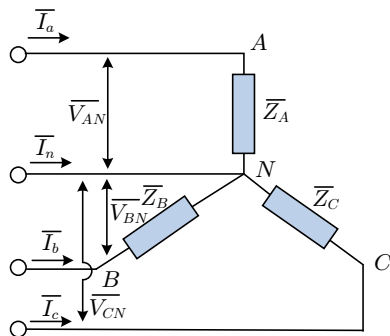


Figure: Unbalanced three-phase Y-connected load

Three Phase Power

$$v_{AN} = \sqrt{2}V_p \cos \omega t, \quad v_{BN} = \sqrt{2}V_p \cos(\omega t - 120^\circ)$$

$$v_{CN} = \sqrt{2}V_p \cos(\omega t + 120^\circ)$$

$$i_a = \sqrt{2}I_p \cos(\omega t - \theta), \quad i_b = \sqrt{2}I_p \cos(\omega t - \theta - 120^\circ)$$

$$i_c = \sqrt{2}I_p \cos(\omega t - \theta + 120^\circ)$$

$$\begin{aligned} P &= P_a + P_b + P_c = v_{AN}i_a + v_{BN}i_b + v_{CN}i_c \\ &= 2V_p I_p [\cos \omega t \cos(\omega t - \theta) \\ &\quad + \cos(\omega t - 120^\circ) \cos(\omega t - \theta - 120^\circ) \\ &\quad + \cos(\omega t + 120^\circ) \cos(\omega t - \theta + 120^\circ)] \end{aligned}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

Three Phase Power (cont..)

$$\begin{aligned}P &= V_p I_p [3\cos\theta + \cos(2\omega t - \theta) + \cos(2\omega t - \theta - 240^\circ) + \cos(2\omega t - \theta + 240^\circ)] \\&= V_p I_p [3\cos\theta + \cos\alpha + \cos\alpha\cos240^\circ + \sin\alpha\sin240^\circ + \cos\alpha\cos240^\circ - \sin\alpha\sin240^\circ] \\&\qquad\qquad\qquad\text{where } \alpha = 2\omega t - \theta \\&= V_p I_p [3\cos\theta + \cos\alpha + 2(-\frac{1}{2})\cos\alpha] \\&= 3V_p I_p \cos\theta\end{aligned}$$

The total instantaneous power in a balanced three-phase system is constant. It does not change with time as the instantaneous power of each phase does. This result is true whether the load is Y or Δ connected.

Three Phase Power (cont..)

Active power per phase is

$$P_p = V_p I_p \cos\theta$$

and the reactive power per phase is

$$Q_p = V_p I_p \sin\theta$$

The apparent power per phase is

$$S_p = V_p I_p$$

The complex power per phase is

$$\bar{S}_p = P_p + jQ_p = \bar{V}_p \bar{I}_p^*$$

where $\bar{V}_p$ and $\bar{I}_p$ are the phase voltage and phase current with magnitudes V_p and I_p respectively.

Three Phase Power (cont..)

The total average power is the sum of the average powers in the phases:

$$P = P_a + P_b + P_c = 3P_p = 3V_p I_p \cos\theta = \sqrt{3}V_L I_L \cos\theta$$

$$Q = 3V_p I_p \sin\theta = 3Q_P = \sqrt{3}V_L I_L \sin\theta$$

and the total complex power is

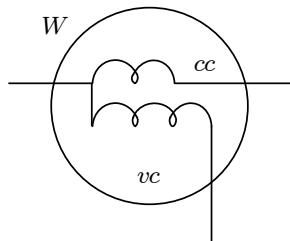
$$\bar{S} = 3\bar{S}_p = 3\bar{V}_p \bar{I}_p^* = 3I_p^2 \bar{Z}_p = \frac{3V_p^2}{\bar{Z}_p^*}$$

where $\bar{Z}_p = Z_p \angle \theta$ is the load impedance per phase. ($\bar{Z}_p$ could be $\bar{Z}_Y$ or $\bar{Z}_\Delta$.) Alternatively,

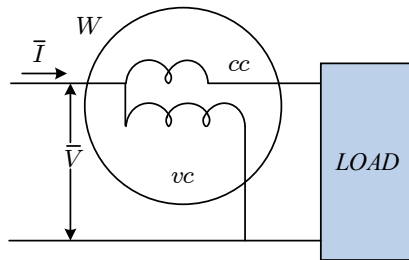
$$\bar{S} = P + jQ = \sqrt{3}V_L I_L \angle \theta$$

Power Measurement

Wattmeter is the instrument used for power measurement. The figure below shows a wattmeter (W) and its connection with a single-phase load. It has two coils, one is called as current coil (cc) and the other is the voltage coil (vc). The current coil is connected in series and the voltage coil is connected across the load or supply whose power is intended to be measured.



(a)



(b)

Two Wattmeter Method

$$\begin{aligned}P_1 &= \operatorname{Re}[\overline{V_{ab}} \overline{I_a}^*] = \overline{V_{ab}} \overline{I_a} \cos(\theta + 30^\circ) \\&= V_L I_L \cos(\theta + 30^\circ)\end{aligned}$$

$$\begin{aligned}P_2 &= \operatorname{Re}[\overline{V_{cb}} \overline{I_c}^*] = \overline{V_{cb}} \overline{I_c} \cos(\theta - 30^\circ) \\&= V_L I_L \cos(\theta - 30^\circ)\end{aligned}$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

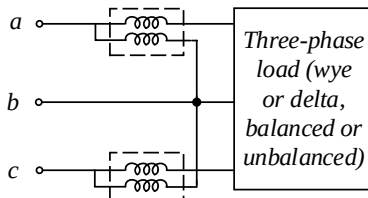


Figure: Two-wattmeter method for measuring three-phase power

Two Wattmeter Method (cont..)

$$\begin{aligned}P_1 + P_2 &= V_L I_L [\cos(\theta + 30^\circ) + \cos(\theta - 30^\circ)] \\&= V_L I_L (\cos\theta \cos 30^\circ - \sin\theta \sin 30^\circ \\&\quad + \cos\theta \cos 30^\circ + \sin\theta \sin 30^\circ) \\&= V_L I_L 2\cos 30^\circ \cos\theta \\&= \sqrt{3} V_L I_L \cos\theta\end{aligned}$$

$$P = P_1 + P_2$$

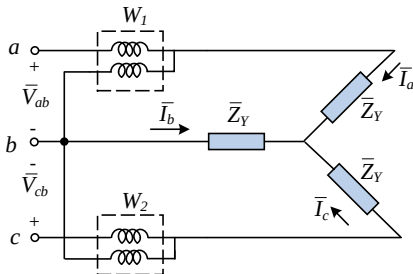


Figure: Two-wattmeter method applied to a balanced wye load

Two Wattmeter Method (cont..)

$$\begin{aligned}P_1 - P_2 &= V_L I_L [\cos(\theta + 30^\circ) - \cos(\theta - 30^\circ)] \\&= V_L I_L (\cos\theta\cos30^\circ - \sin\theta\sin30^\circ - \cos\theta\cos30^\circ - \sin\theta\sin30^\circ) \\&= -V_L I_L 2\sin30^\circ\sin\theta \\&= -V_L I_L \sin\theta\end{aligned}$$

$$Q = \sqrt{3}(P_2 - P_1)$$

$$S = \sqrt{P^2 + Q^2}$$

$$\tan\theta = \frac{Q}{P} = \sqrt{3} \frac{P_2 - P_1}{P_1 + P_2}$$

Two Wattmeter Method (cont..)

1. If $P_2 = P_1$, the load is resistive.
2. If $P_2 > P_1$, the load is inductive.
3. If $P_2 < P_1$, the load is capacitive.

Thank You!